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Acoustical modeling of micro-perforated panel at high sound pressure levels using equivalent fluid approach

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ABSTRACT

An acoustic impedance model to characterize micro-perforated panels at high sound pressure levels is proposed. The model uses a rigid frame porous medium, where the micro-perforated panel is modeled with an effective density, function of the frequency following the approach of Johnson-Allard and the equivalent fluid parameters such as the tortuosity and the flow resistivity are expressed as function of the incident sound pressure which is considered as a main variable. Unlike existing models which are limited to micro-perforated panels coupled to air cavity, the present model predicts correctly the acoustic response of micro-perforated panel backed by porous media which can be air cavity, porous material or resistive screen. Micro-perforated panel backed by porous media involves a distortion of the flow caused by the perforations through the porous media, thus the micro-perforated panel is modeled in this case using an equivalent tortuosity where a correction term is proposed to account for the flow distortion effect and it depends on the dynamic tortuosity of the porous layer and the incident sound pressure. The proposed impedance model is compared numerically with other existing nonlinear impedance models for different configurations of the micro-perforated panel. The results show a good agreement among each other for sound pressure levels up to 150 dB. In addition, experimental measurements were performed on several micro-perforated panels backed by air cavities or porous material using the classical impedance tube. A good correlation between theoretical and experimental results is obtained. Some validation and benchmarking results are illustrated and discussed in this paper. It is shown that the high sound pressure levels decrease the tortuosity and increase the flow resistivity of the micro-perforated panel. Furthermore, the acoustic energy dissipation by the micro-perforated panel absorber with a large perforation diameter which is low in the linear regime can become important and interesting at high sound pressure level.

1. Introduction

Micro-perforated panel (MPP) unlike ordinary perforated panel where the perforations are in order of millimeters or even centimeters is a panel where the perforations are reduced to submillimeter size to generate more acoustic resistance with a low perforation ratio and low acoustic mass reactance. In many application fields such as civil engineering, exhaust systems

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and ducts, architecture, transportation industry and aeronautic, MPP is widely used for noise reduction. In turbofan engine, the nacelle is lined with the acoustic liners made of a thin layer (MPP or wire-mesh), a cellular separator such as honeycomb and a rigid backplate in order to reduce the engine fan noise. Conventional porous materials like foams and fibres are flammable and difficult to maintain and cannot be used in environments with a temperature gradient while the MPP can withstand high temperatures. The MPP absorber presents an interesting sound absorption in low and medium frequency.

In linear regime, different theoretical and experimental approaches were carried out to characterize the acoustic properties of the MPP. Atalla and Sgard [1] and Atalla et al. [2] modeled the MPP as an equivalent fluid using Johnson-Allard method [3] for a rigid frame porous material. Maa [5] developed an acoustic impedance model of MPP based on Crandall's theory [6] for the acoustic wave propagation in short and narrow circular tubes. The absorbent system constituted of MPP backed by an air cavity and rigid wall can be assimilated to a distribution of Helmholtz resonator [3] with viscous and thermal dissipation on the surfaces and the neck. The effect of interaction between perforations on the acoustic properties of the MPP absorber is analysed by some authors [7,13]. The sound absorption mechanism in the linear regime is the conversion of acoustic energy into heat through friction by viscous and thermal effects when the dimensions of the perforations are in the order of viscous and thermal boundary layers thicknesses.

Several studies [4,8–21,28] were published on the acoustic nonlinear effects of orifices. Maa [8] explained that the acoustic non linearity of an orifice is an external phenomenon and proposed a nonlinear acoustic impedance model of the MPP where the specific nonlinear resistance expression is the Mach number in the perforation of the MPP divided by the percentage open area (POA). Ingard and Labate [9] noted the jet formation and vortex at the exit of orifices by investigating circulations in air caused by sound waves. They showed the existence of pulsatory effects resulting in the formation of jets and vortex rings at high sound pressure levels (SPL). The nonlinear properties of the acoustic impedance of an orifice are closely connected with the circulation effects. Ingard and Ising [10] investigated experimentally the nonlinear acoustic behavior of an orifice by measuring simultaneously with a hot wire the flow velocity in the orifice and the acoustic pressure fluctuations. In the linear regime, they showed that the relation between the pressure and velocity amplitudes was linear but at large velocity amplitude, it approached a square-law relation and in this region, the acoustic resistance of the orifice dominated and varied linearly with respect to the orifice particle velocity. Furthermore, they demonstrated by their measurements at high SPL the flow separation through the orifice in the form of a high velocity jet. Ingard [11] computed the nonlinear absorption characteristics of a resonator absorber as a function of frequency. He proposed a nonlinear resistance of an orifice which is dependent upon the acoustic particle velocity in the orifice. Guess [12] determined the parameters of a single layer perforated plate by a mathematical procedure leading to elaboration of a specific acoustic resistance and reactance of MPP. Melling [13] developed a theoretical modeling of the impedance of perforates at higher levels based on a quasi-steady approximation to the acoustic flow through the orifice. This approach yields a formula for the nonlinear acoustic resistance that depends on the discharge coefficient. Chang and Cummings [14] and Cummings [15,16] studied the acoustic behavior of orifices at high SPL using a time domain approach. They presented one dimensional numerical model for the perforated plate. Kraft et al. [17,18] worked on acoustic treatment panels for high SPL leading to development of an impedance model of MPP where the resistance and the reactance are independent of frequency. Hersh et al. [19] derived one dimensional empirical impedance model for Helmholtz resonators constructed with circular orifices. The resonator geometry and incident sound pressure amplitude are considered in the model. Tayong et al. [20] proposed a model based on dimensional analysis and Forchheimer's law to investigate the acoustic behavior of MPP at high pressure excitation. Recently, Soon-Hong PARK [21] developed an empirical acoustic resistance model of a MPP which is given as a function of the incident sound pressure and the parameters of the MPP such as the thickness of the panel, the diameter of the orifice and the perforation ratio.

In this paper, an acoustic impedance model for predicting the acoustic response of MPP in high sound pressure environment is presented in section 3. The model is an extension of the linear equivalent fluid impedance model developed by Atalla and Sgard [1]. The parameters of the linear impedance model are modified to account for the nonlinear phenomena induced by the high SPL. The flow resistivity and the tortuosity of the MPP are shown to be dependent upon the acoustic particle velocity in the orifice and are expressed as functions of the incident pressure on the surface of the perforations. The main variable in the model is the incident pressure. The proposed model is verified by comparison with others nonlinear existing models and experimental results in sections 4 and 5. In section 6, micro-perforated panels backed by porous media are modeled and tested experimentally. Unlike existing models which are limited to MPP backed by an air cavity, the present model predicts well the acoustic properties of MPP at high SPL and couples with various domains such as an air cavity, a porous material or a resistive screen. When a porous media backs the MPP, a correction to the equivalent tortuosity is proposed to account for the effect of the flow distortion caused by the perforations through the porous media. Finally, section 7 presents the effect of the high SPL on the acoustic behavior of the MPP absorber. It is demonstrated that the increase in SPL causes a decrease of the tortuosity while the flow resistivity increases and the nonlinear phenomena dissipate acoustic energy so that a MPP absorber with a large orifice diameter which is not efficient at low pressure levels can be a good absorber at high SPL.

2. Review of acoustic impedance models of MPP at high SPL

Cummings [15] noted that the loss of acoustic power at the orifice is consistent with the transfer of this power into the kinetic energy of two trains of ring vortices shed alternately from both sides of the orifice. Ingard and Labate [9] explained that the nonlinear absorption of a sound wave is due to the presence of circulation effects; in the jet region, the acoustic energy is transformed into kinetic energy of the jet which should equal the nonlinear part of the absorbed energy in the sound wave. Several studies [8,12,13,21] show that the normalized specific resistance of the MPP results from the addition of the linear and nonlinear resistance terms and is written as

$$R = R_L + \theta_{nl}, \tag{1}$$

where R_L is the normalized specific linear resistance and θ_{nl} is the normalized nonlinear specific resistance. The results of Ingard and Ising [10] indicate that the orifice resistance varies linearly with the velocity amplitude when the orifice diameter is larger than the panel thickness. The specific nonlinear resistance of the MPP due to the high SPL is given according to Ingard [11] by

$$\theta_{nl} = \frac{(1-\phi^2)}{\phi c_0} V_a, \tag{2}$$

with ϕ the Percentage Open Area (POA) of the MPP, c_0 the speed of sound in air and V_a the amplitude of the particle acoustic velocity in the orifice. Zinn [26] proposed a nonlinear resistance of Helmholtz resonator as a function of discharge coefficient with an empirical factor $4/3\pi$ and as reported in others works [12,13,30], the nonlinear resistance θ_{nl} can be given by

$$\theta_{nl} = \frac{4}{3\pi} \frac{(1-\phi^2)}{\phi c_0 C_D^2} V_a, \tag{3}$$

where C_D is the discharge coefficient varying between 0.6 and 0.8. The expression of the resistance in Eq. (3) was used by Malmary et al. [29] to compare impedance models of MPP and experimental measurements with grazing air flow. The high excitation pressure involves nonlinear phenomena which change also the reactance of the perforated panel. It was suggested by Ingard and Ising [10] that nonlinear orifice jetting reduces orifice end correction. Furthermore, experimental measurements in Ref. [27] show that as the SPL is increased, the end correction approaches one-half of its low amplitude value [12]. This is due to the turbulent jet formation at the exit of the orifice. Thus, Guess [12] proposed to multiply the correction length in the expression of the linear acoustic reactance of the MPP by the following factor

$$\xi = \frac{1 + 5000 M_a^2}{1 + 10000 M_a^2}. \tag{4}$$

The quantity $M_a = V_a/c_0$ is the orifice Mach number.

Fig. 1 illustrates MPP backed by an air cavity and a rigid wall with the sound incidence pressure on the surface of the perforations $\overline{P}_i$. The hole diameter is d, the thickness of the panel is h and the cavity depth is D. The system is known as micro-perforated panel absorber. It is a locally reacting absorber where sound propagation is normal to the MPP surface. The impedance of a locally reacting absorber is independent of the angle of incidence [23].

The acoustical propagation in short tube [22] was studied by Crandall [6]. Based on this theory, the normalized acoustic impedance of an orifice in the linear regime was derived as follows [13]:

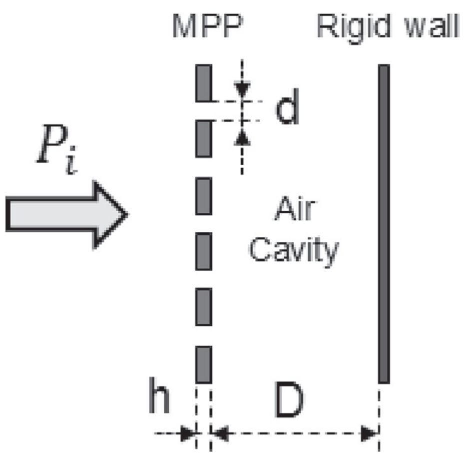

Fig. 1. Micro-perforated panel backed by air cavity.

$$Z_{perf} = j\omega \frac{h}{c_0}\left(1 - \frac{2}{x\sqrt{-j}}\frac{J_1\left(x\sqrt{-j}\right)}{J_0\left(x\sqrt{-j}\right)}\right)^{-1}, \tag{5}$$

where ω is the angular frequency, j is the imaginary complex number ($j^2 = -1$), J_0 and J_1 are respectively the Bessel function of the first kind of orders 0 and 1, the perforate constant x is proportional to the ratio of the radius to the viscous boundary layer thickness of the air in the orifice and is given by $x = 0.5d\sqrt{\omega\rho_0/\eta}$, with η the dynamic viscosity and ρ_0 the density of air. The limiting values of Z_{perf} in Eq. (5) based on Bessel functions are [6] $4j\rho_0 h\omega/3 + 32\eta h/d^2$ as $x<1$ and $j\rho_0 h\omega + 4\eta h(1+j)\sqrt{\rho_0\omega/2\eta}/d$ as $x>10$. The acoustic impedance of a perforation is linked to the impedance of the MPP by the POA ϕ. An end correction is taken into account for the impedance due to the sound radiating effects at the aperture of the perforation. This effect is identified to the radiation of a circular piston in a baffle [24]. The reactance of the perforation is corrected by adding the reactance of radiation impedance of the piston which is multiplied by two for both ends. The vibration of air particles in the vicinity of the aperture on the baffle causes an additional viscosity that is accounted for by Rayleigh [25] who proposed an additional factor to the resistance given by $R_s = 0.5\sqrt{2\eta\rho_0\omega}$. Using Eqs. (3)–(5), the normalized impedance of MPP at high SPL can then be written as

$$\begin{aligned} Z_{NL} = {} & \frac{\sqrt{2}\eta x}{\rho_0 c_0 \phi d} + \frac{\omega^2 d^2}{8\phi c_0^2} + \frac{4(1-\phi^2)}{3\pi\phi C_D^2}\frac{V_a}{c_0} \\ & + \frac{j\omega}{\phi c_0}\left\{\frac{8d}{3\pi}\xi + h\left[1 - \frac{2}{x\sqrt{-j}}\frac{J_1\left(x\sqrt{-j}\right)}{J_0\left(x\sqrt{-j}\right)}\right]^{-1}\right\}. \end{aligned} \tag{6}$$

where ξ is given in Eq. (4). In Eq. (6), $\frac{\sqrt{2}\eta x}{\rho_0 c_0 \phi d}$ corresponds to the resistance caused by the vibration of air particles in the vicinity of the orifice and $\frac{\omega^2 d^2}{8\phi c_0^2} + j\frac{8\omega d}{3\pi\phi c_0}$ results from the radiation impedance of a circular piston in a baffle [24].

Maa [8] studied the high SPL effects on the acoustic impedance of MPP. He suggested a specific nonlinear resistance of MPP in terms of March number in the orifice divided by the percentage open area. He accounted for the effects of nonlinear phenomena on the reactance by multiplying the correction length by the factor $1/(1+V_a/\phi c_0)$. Thus, in this model, the normalized resistance R and reactance χ of the MPP are

$$R = \frac{32\eta h}{\rho_0 c_0\, \phi d^2}\left[\sqrt{1+\frac{x^2}{32}} + \frac{\sqrt{2}xd}{32h}\right] + \frac{V_a}{\phi c_0}, \tag{7}$$

$$\chi = \frac{\omega h}{\phi c_0}\left[1 + \frac{1}{\sqrt{9+x^2/2}} + 0.85\frac{d}{h}\left(1+\frac{V_a}{\phi c_0}\right)^{-1}\right]. \tag{8}$$

Eqs. (7) and (8) are derived from the limiting values of Eq. (5) and they are valid when $1<x<10$ [6,21]. The nonlinear normalized resistance of a panel with a perforation can be expressed by $V_a(1-\phi^2)/\phi c_0$ using Bernoulli's law [21]. For a low perforation ratio, this nonlinear resistance term becomes $V_a/\phi c_0$; this is the expression used in Maa's model. The POA should in consequence be low in order to use this model. Moreover, Maa's nonlinear resistance term depends only on the POA and does not take into account the other parameters of the MPP such as the perforation diameter and the plate thickness. Note that a large discrepancy was observed by Soon-Hong PARK [21] between the predicted resistances using Maa's model and his measurements results.

Soon-Hong PARK [21] proposed an empirical nonlinear resistance θ_{nl} of MPP by considering all geometric parameters of MPP. After an experimental investigation, the nonlinear resistance is given by

$$\theta_{nl} = 1.59\phi^{-0.845}\left(\frac{d}{h}\right)^{0.06}\frac{V_a}{c_0}. \tag{9}$$

This model accounts for the parameters of the MPP and is appropriate for MPP in the launcher fairing application where the overall SPL is around 140 dB. The results of this model are in good agreement with measurements when the perforation ratio is in the range of 1.4%–5.3%. There is no guarantee for the validity of this model outside this range of perforation ratio.

3. The proposed impedance model

In the linear regime, Atalla and Sgard [1] developed an impedance model of a MPP or resistive screens based on an equivalent fluid following the Johnson-Allard approach [3]. The normalized acoustic impedance of the MPP is written as

$$Z_{Linear} = j\frac{\omega h}{\rho_0 c_0\ \phi}\tilde{\rho}_e. \tag{10}$$

The effective density $\tilde{\rho}_e$ depends of the parameters of the MPP and the frequency. It is linked to the dynamic tortuosity $\tilde{\alpha}$ by the air density ρ_0 ($\tilde{\rho}_e(\omega) = \tilde{\alpha}(\omega)\rho_0$) and is given by

$$\tilde{\rho}_e(\omega) = \alpha_\infty \rho_0 \left(1 + \frac{\sigma\phi}{j\omega\alpha_\infty\rho_0}\sqrt{1 + \frac{4j\rho_0\omega\eta\alpha_\infty^2}{\phi^2\sigma^2\Lambda^2}}\right). \tag{11}$$

In the case of cylindrical orifices, the flow resistivity σ is related to the POA ϕ and the radius of the perforation r of the MPP by $\sigma = 8\eta/(\phi r^2)$, the viscous characteristic length Λ is equal to the hydraulic radius of the perforation ($\Lambda = r$) and the tortuosity α_∞ is given by

$$\alpha_\infty = 1 + \frac{2\varepsilon_e}{h}, \tag{12}$$

where ε_e is the correction length which is a function of the perforation radius r and the percentage open area, $\varepsilon_e = 0.48\sqrt{\pi r^2}(1 - 1.14\sqrt{\phi})$ with $\sqrt{\phi} < 0.4$. This model is valid for low SPL. If one increases the SPL, the nonlinear phenomena appear, changing the parameters (the flow resistivity σ and the tortuosity α_∞) of the model. Thus this model cannot be used to characterize the acoustic behavior of MPP in high sound pressure environment. In the present work, the flow resistivity σ and the tortuosity α_∞ are modified to extend the model to high SPL. The nonlinear phenomena reduce the correction length [27] and because the tortuosity of the MPP is a function of the correction length, it will be directly affected by the nonlinear phenomena and will depend upon the particle velocity in the orifice. The flow resistivity will be derived from the specific nonlinear resistance of the MPP which increases significantly with the SPL.

The linear impedance model in Eq. (10) can be simplified according to the high or low frequencies. At high frequencies ($\omega D/c_0 > > 1$), the expression of the dynamic tortuosity is [3].

$$\lim_{\omega\to\infty} \tilde{\alpha}(\omega) = \alpha_\infty \left(1 + \frac{(1-j)}{\Lambda}\sqrt{\frac{2\eta}{\omega\rho_0}}\right) \tag{13}$$

The effective density $\tilde{\rho}_e$ becomes

$$\tilde{\rho}_e(\omega) = \alpha_\infty \rho_0 \left(1 + \frac{(1-j)}{\Lambda}\sqrt{\frac{2\eta}{\omega\rho_0}}\right) \tag{14}$$

Substituting Eq. (14) into Eq. (10) yields

$$Z_{Linear} = j\frac{\omega h}{\phi c_0}\alpha_\infty\left(1 + \sqrt{\frac{2\eta}{\rho_0\omega\Lambda^2}}\right) + \frac{h\alpha_\infty}{\rho_0 c_0\ \phi\Lambda}\sqrt{2\eta\rho_0\omega}. \tag{15}$$

Eq. (15) shows that the acoustic impedance of the MPP for high frequencies is independent of the flow resistivity. The phenomenon that governs the impedance is the inertial effect that amounts to increasing the mass of the vibrating air and is accounted by using the correction length [1]. The low frequency expression of the dynamic tortuosity is obtained for $\omega D/c_0 < < 1$; it is given by Ref. [3].

$$\lim_{\omega\to 0} \tilde{\alpha}(\omega) = \alpha_\infty + \frac{\sigma\phi}{j\omega\rho_0} + \frac{2\alpha_\infty^2\eta}{\sigma\phi\Lambda^2} \tag{16}$$

Thus Eq. (10) can be written using Eq. (16) as

$$Z_{Linear} = j\frac{\omega h}{\phi c_0}\alpha_\infty\left(1 + \frac{2\eta\alpha_\infty}{\phi\sigma\Lambda^2}\right) + \frac{\sigma h}{\rho_0 c_0}. \tag{17}$$

As it is known that the resistance of the MPP is the contribution of two quantities namely the linear and nonlinear resistance, one can then deduce from Eq. (17) that the flow resistivity of the MPP for high SPL results from the contribution of two terms that are the linear flow resistivity σ used by Atalla and Sgard [1] and the nonlinear flow resistivity σ_{nl} caused by the increase in SPL. By dividing the resistive part by the thickness of the MPP, the flow resistivity will be found at low frequencies. The proposed expression of the flow resistivity σ_t of MPP in higher pressure environment is

$$\sigma_t = \sigma + \sigma_{nl}. \tag{18}$$

The nonlinear flow resistivity σ_{nl} is derived from the nonlinear resistance θ_{nl} of MPP. Thus, the general expression of the flow resistivity is written as follows:

$$\sigma_t = \sigma + \zeta \rho_0 c_0 \frac{\theta_{nl}}{h}, \tag{19}$$

where ζ is a constant value.

Using the nonlinear resistance θ_{nl} in Eq. (3), the flow resistivity σ_t at high SPL can be expressed as

$$\sigma_t = \frac{8\eta}{\phi r^2} + \beta \frac{\rho_0 (1-\phi^2)}{\pi h \phi C_D^2} V_a, \tag{20}$$

where β is a constant value.

The tortuosity of the MPP is a function of the correction length that accounts for the sound radiation effect at the end of the perforations. As mentioned before, the turbulent jet formation on the side of the perforation reduces the orifice end correction. Thus, the tortuosity will decrease when the SPL becomes higher. In the proposed expression for the tortuosity, one multiplies the correction length ε_e in the linear impedance model [1] by the factor $1/(1+V_a/\phi c_0)$ following Maa [8]. This factor is a decreasing function of the particle velocity V_a in the orifice. Thus, the proposed expression of the tortuosity at high pressure level is given by

$$\alpha_{\infty nl} = 1 + \frac{2\varepsilon_{enl}}{h}. \tag{21}$$

In Eq. (21), ε_{enl} denotes the correction length accounting for the nonlinear effects and interaction between perforations. It is given by

$$\varepsilon_{enl} = \frac{\Psi}{(1+V_a/(\phi c_0))} 0.48\sqrt{\pi r^2}\left[\sum_{n=0}^{8} a_n \left(\sqrt{\phi}\right)^n\right], \tag{22}$$

where Ψ is a constant value set to 4/3, the parameters a_n are given by

$$a_0 = 1.0\ ,\ a_1 = -1.4092\ ,\ a_2 = 0.0\ ,\ a_2 = 0.33818\ ,\ a_4 = 0.0$$
$$a_5 = 0.06793\ ,\ a_6 = -0.02287\ ,\ a_7 = 0.003015\ ,\ a_8 = -0.01614$$

The effective density of the MPP at high SPL can now be determined by using the expressions of the flow resistivity σ_t in Eq. (20) and the tortuosity $\alpha_{\infty nl}$ in Eq. (21)

$$\tilde{\rho}_{en}(\omega) = \rho_0 \alpha_{\infty nl}\left(1 + \frac{\sigma_t \phi}{j\omega\rho_0\alpha_{\infty nl}}\sqrt{1 + \frac{4j\rho_0\omega\eta\alpha_{\infty nl}^2}{\phi^2\sigma_t^2\Lambda^2}}\right). \tag{23}$$

The effective density $\tilde{\rho}_{en}$ in Eq. (23) will vary with respect to the particle velocity V_a depending upon the sound pressure. The normalized acoustic impedance of the MPP at high SPL is then given by

$$Z_{MPP} = j\frac{\omega h}{\rho_0 c_0\ \phi}\tilde{\rho}_{en}. \tag{24}$$

The use of Eqs. (20) and (22) necessitates an estimation of the velocity in the orifice. In practice, the calculation of the root-mean-squared (rms) velocity $\overline{V}_a$ in the orfice is done iteratively, since it depends on the unknown impedance of the liner. For a sound pressure level L_P at the surface of the facesheet, the rms velocity can be obtained by the following equation [38].

$$\overline{V}_a = \frac{P_{ref}10^{L_P/20}}{\rho_0 c_0 |Z|}, \tag{25}$$

where $|Z|$ is the normalized impedance of the sample under test and P_{ref} the reference pressure ($P_{ref} = 20\ \mu$ Pa). The impedance model in Eq. (24) can be computed by an iteration procedure [38]. Starting from an initial guess of $\overline{V}_a$, one calculates the flow resistivity using Eq. (20) and the tortuosity using Eq. (21), then use Eq. (24) to estimate the impedance and use it in Eq. (25) for a new value of $\overline{V}_a$. This procedure must be repeated until the impedance converges. An alternative method to estimate $\overline{V}_a$ is given by Soon-Hong PARK [21]. The relation between the rms velocity $\overline{V}_a$ in the orifice and the rms incident pressure $\overline{P}_i$ can be obtained by using the acoustic circuit analogy [34] and the momentum equation in the form of Bernouilli's law for a laminar and incompressible fluid. The rms velocity is found to be [21].

$$\overline{V}_a = \frac{c_0}{\sqrt{2}}\frac{\phi}{(1-\phi^2)}\left[-\frac{1}{2}+\sqrt{\frac{1}{4}+\frac{2\sqrt{2}\overline{P}_i}{\rho_0 c_0^2}\frac{(1-\phi^2)}{\phi^2}}\right]. \tag{26}$$

The flow resistivity of the MPP (Eq. (20)) can be expressed as function of the rms incident pressure $\overline{P}_i$ which is the main variable

$$\sigma_t = \frac{8\eta}{\phi r^2} + \beta\frac{\rho_0 c_0}{\pi h C_D^2}\left[-\frac{1}{2}+\sqrt{\frac{1}{4}+\frac{2\sqrt{2}\overline{P}_i}{\rho_0 c_0^2}\frac{(1-\phi^2)}{\phi^2}}\right]. \tag{27}$$

Similarly, the tortuosity (Eq. (21)) is expressed as a function of the incident pressure $\overline{P}_i$

$$\alpha_{\infty nl} = 1 + \frac{2\Psi}{h}0.48\sqrt{\pi r^2}\left[\sum_{n=0}^{8} a_n\left(\sqrt{\phi}\right)^n\right]\left[1+\frac{1}{(1-\phi^2)}\left(-\frac{1}{2}+\sqrt{\frac{1}{4}+\frac{2\sqrt{2}\overline{P}_i}{\rho_0 c_0^2}\frac{(1-\phi^2)}{\phi^2}}\right)\right]^{-1}. \tag{28}$$

The effective density of the MPP in Eq. (23) can now be determined as function of the incident pressure $\overline{P}_i$ by using the flow resistivity and the tortuosity which are given in Eqs. (27) and (28). The acoustic nonlinear impedance of the MPP in Eq. (24) can then be expressed as function of the incident pressure $\overline{P}_i$ which is calculated by the relation $\overline{P}_i = P_{ref}10^{L_{P_i}/20}$ with L_{P_i} the incident SPL.

The resonant sound absorption frequency of the MPP absorber is calculated using the low frequency expression (Eq. (17)) of the surface impedance which is expressed in the nonlinear regime as

$$Z_{MPP} = j\frac{\omega h}{\phi c_0}\alpha_{\infty nl}\left(1+\frac{2\eta\alpha_{\infty nl}}{\phi\sigma_t\Lambda^2}\right) + \frac{\sigma_t h}{\rho_0 c_0} - j\cot\left(\frac{\omega D}{c_0}\right). \tag{29}$$

Eq. (29) is valid for low frequencies range ($\omega D/c_0 < < 1$) with $\sqrt{\phi} < 0.4$.

The resonances of the MPP absorber where the sound absorption coefficient is maximum occur at zeros of the reactance term

$$\frac{\omega h}{\phi c_0}\alpha_{\infty nl}\left(1+\frac{2\eta\alpha_{\infty nl}}{\phi\sigma_t\Lambda^2}\right) = \cot g\left(\frac{\omega D}{c_0}\right). \tag{30}$$

At low frequencies, the first mode (resonance) of the MPP absorber is given by

$$\omega_r = \sqrt{\frac{\phi c_0^2}{Dh\alpha_{\infty nl}\left(1+\frac{2\eta\alpha_{\infty nl}}{\phi\sigma_t\Lambda^2}\right)}}. \tag{31}$$

Fig. 2 illustrates the effect of the cavity depth at 140 dB on the resonant sound absorption frequency (Eq. (31)) for a MPP absorber with thickness of 1.0 mm, the hole diameter is 0.3 mm and the perforation ratio is 3.0%. The resonant frequency decreases from 3500 Hz to 1000 Hz when the cavity depth increases from 10 mm to 50 mm. Thus a low frequency design of the MPP absorber requires a larger cavity depth because the wavelengths are large in this frequency range. However, the absorption coefficient peak decreases with the cavity depth. This is normal since a MPP backed by an air cavity and rigid wall

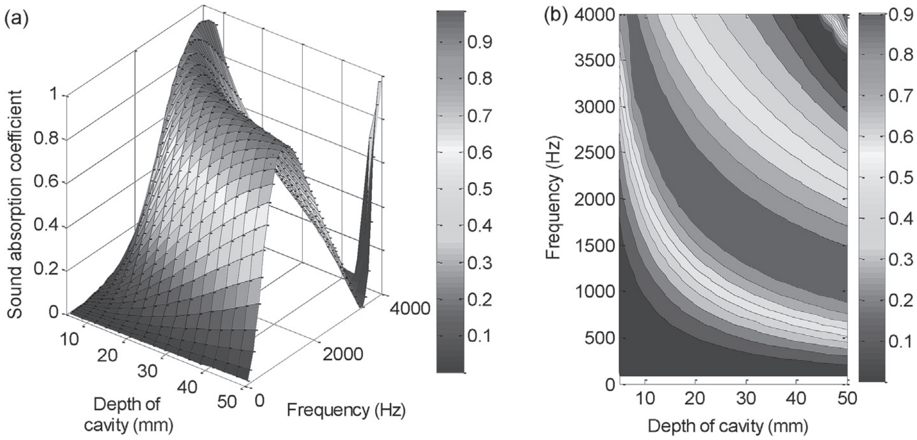

Fig. 2. Normal incidence sound absorption coefficient of MPP absorber (SPL = 140 dB, thickness = 1.0 mm, hole diameter = 1.0 mm, POA = 6%): (a) surface plot, (b) contour plot.

is mainly a distribution of Helmholtz resonators. The associated quality factor Q is defined by Ref. [41] $Q = \omega\left(\frac{\text{Energy stored}}{\text{Power dissipated}}\right)$. If l_n and S_n are respectively the effective length and the cross-sectional area of the neck and V_R the volume of the air cavity, the quality factor is given by $Q = 2\pi\sqrt{l_R^3 V_R / S_R^3}$ [39]. Thus if one increases the cavity depth; the volume of the cavity increases and the quality factor increases. This implies a low acoustic dissipation [41].

4. Comparison with existing models

The normal incidence sound absorption coefficients estimated by the proposed model (Eq. (24)) are compared with others existing nonlinear impedance models in Figs. 3–6 for a sound pressure level varying between 110 dB and 150 dB and for different parameters of the MPP. The typical value of the discharge coefficient is $C_D \approx 0.76$ [32]. The constant value β is set to 1.6. Figs. 3–6 illustrate that the present proposed model is in good agreement with others existing impedance models.

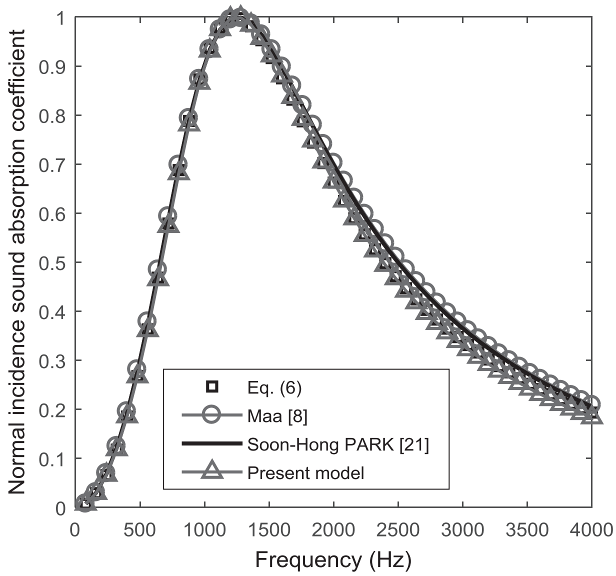

Fig. 3. Normal incidence sound absorption coefficient of MPP absorber (SPL = 110 dB, thickness = 1.0 mm, hole diameter = 0.25 mm, POA = 2.8%, depth of cavity = 30 mm).

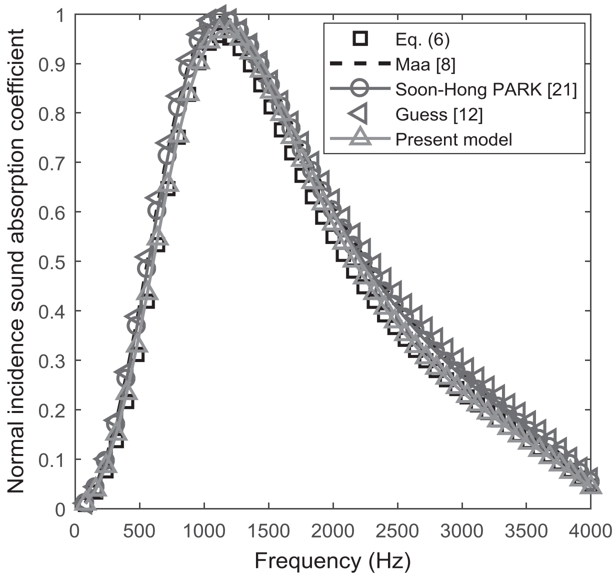

Fig. 4. Normal incidence sound absorption coefficient of MPP absorber (SPL = 135 dB, thickness = 1.2 mm, hole diameter = 1.0 mm, POA = 4.17%, depth of cavity = 40 mm).

5. Validation with measurements

5.1. Comparison of the model with literature data

In this section the proposed impedance model is verified with measurements. First, the comparison is made with published data. Next, new tests performed in our lab (GAUS) on different samples will be used for the comparison.

In Figs. 7 and 8, the estimated normal incidence absorption coefficients by the present impedance model in Eq. (24) are compared with the measured results performed by Soon-Hong PARK [21] on a MPP absorber with thickness of 1.0 mm, the hole diameter is 1.0 mm, the perforation ratio is 5.14% and the cavity depth is 100 mm. The tests are performed at an overall SPL of 115 dB (Fig. 7) and 143 dB (Fig. 8) using an impedance tube.

It is observed that the estimated absorption coefficients by the present impedance model and the measured results are in good agreement. The sharp absorption peak at 790 Hz in Fig. 7 is caused by the vibration of the panel coupled with air cavity [35,36]. This absorption peak was also observed by Tayong et al. [20] in their experimental results.

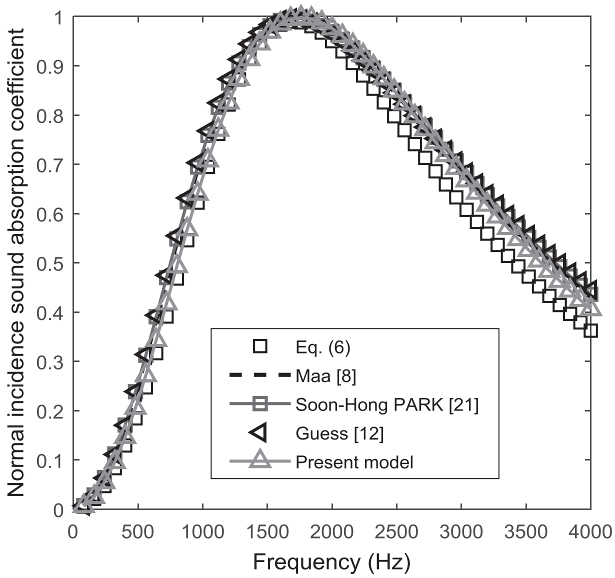

Fig. 5. Normal incidence sound absorption coefficient of MPP absorber (SPL = 143 dB, thickness = 0.8 mm, hole diameter = 1.2 mm, POA = 5.23%, depth of cavity = 28 mm).

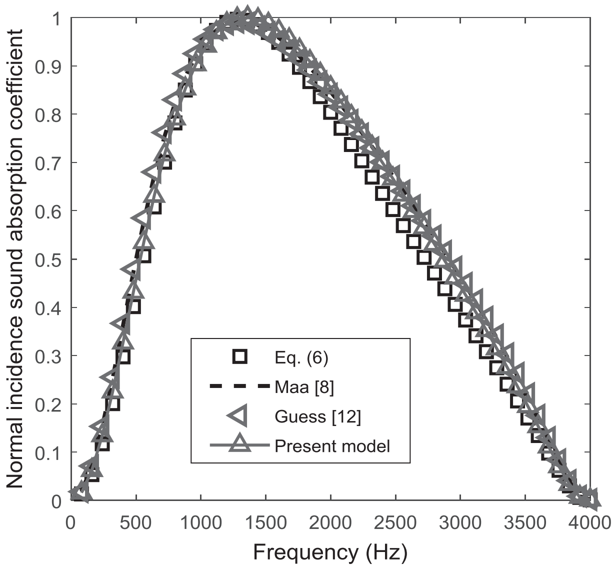

Fig. 6. Normal incidence sound absorption coefficient of MPP absorber (SPL = 150 dB, thickness = 1.2 mm, hole diameter = 1.2 mm, POA = 7.2%, depth of cavity = 43 mm).

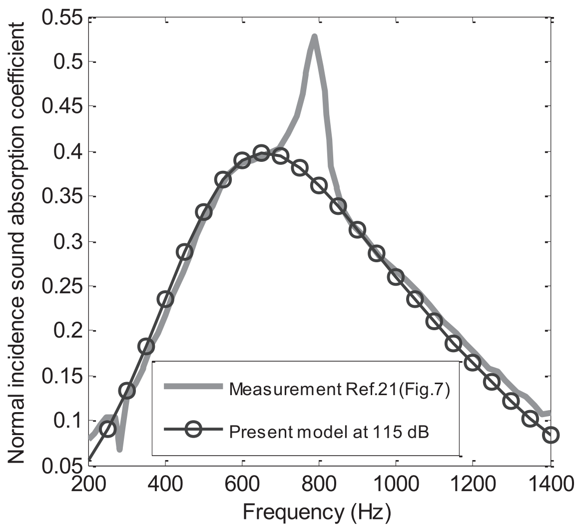

Fig. 7. Normal incidence sound absorption coefficient comparison of MPP absorber at 115 dB (thickness = 1.0 mm, hole diameter = 1.0 mm, POA = 5.14%, depth of cavity = 100 mm).

Fig. 9 compares the normalized specific resistance and reactance of another MPP absorber predicted by the present model and experimental results performed by Kraft et al. [31] in normal incidence impedance tube with broadband excitation at 144.3 dB. The thickness of the MPP is 0.3 mm, the orifice diameter is 0.5 mm, the perforation ratio is 8% and the depth of the back cavity is 12.7 mm.

Again, the predicted normalized specific resistance and reactance of the MPP absorber in Fig. 9 are in good agreement with the experimental results.

5.2. Validation with own measurements

Next, comparisons with our measurements are presented. The tests are done using a high SPL impedance tube. A schematic diagram of the tube is given in Fig. 10. The diameter of the tube is 29 mm (the cut-off frequency is about 6907 Hz). Two 1/4″ microphones (PCB micros 377B10 with 0.89 mV/Pa and 0.90 mV/Pa sensitivities respectively) are mounted flush with the inner wall of the tube and are used to calculate the surface impedance of the sample by the two-microphone standing waves

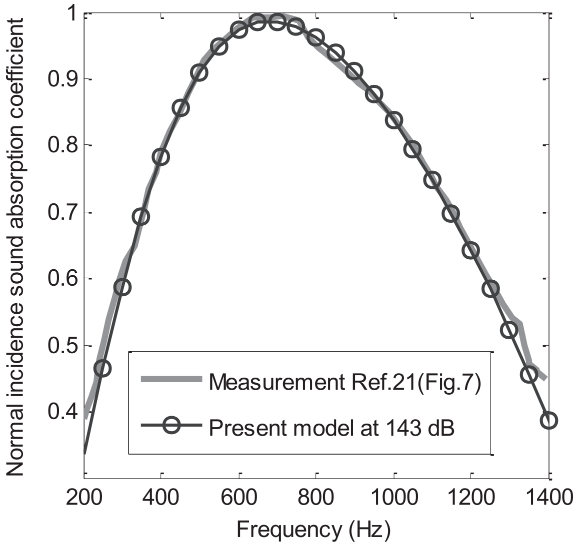

Fig. 8. Normal incidence sound absorption coefficient comparison of MPP absorber at 143 dB (thickness = 1.0 mm, hole diameter = 1.0 mm, POA = 5.14%, depth of cavity = 100 mm).

method [33]. A phase and amplitude calibration is done to correct the transfer function between the measurement microphones. The high sound source of excitation is a compression chamber BMS 4592 mounted at the left hand side of the tube and powered by a power amplifier Stanton A.2800. The test sample is placed in a sample holder and mounted at the right hand side of the tube. A rigid plunger with an adjustable depth is placed behind the sample to create an air cavity. The distance between the two microphones is 20 mm and the distance between the second microphone (micro 2) and the sample is 12.8 mm. The transfer function between the two microphones, specially calibrated to minimise amplitude and phase errors between the channels is obtained with a multi-channel spectrum analyzer.

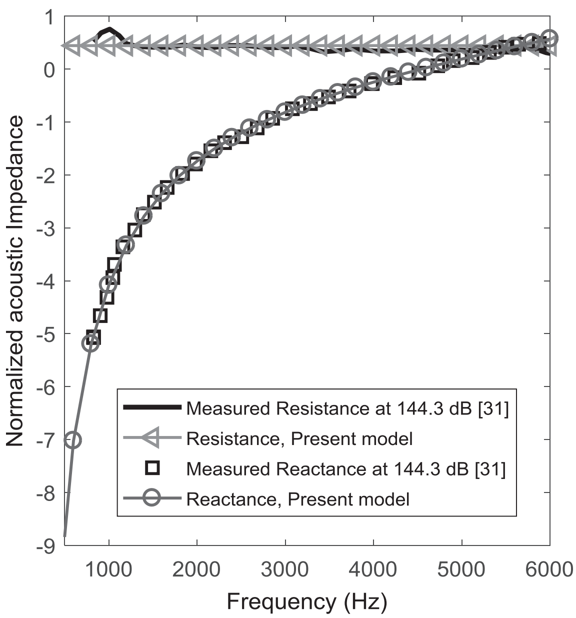

Fig. 9. Normalized surface impedance of MPP absorber at 144.3 dB (thickness = 0.305 mm, hole diameter = 0.508 mm, POA = 8%, depth of cavity = 12.7 mm).

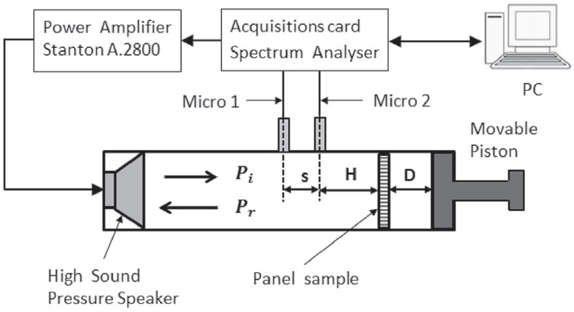

Fig. 10. Schematic diagram of the impedance tube.

Table 1
Geometric parameters of the single MPP absorber for the measurement.

	Thickness (mm)	Hole diameter (mm)	Perforation ratio (%)	Cavity depth (mm)
MPP#1	0.86	1.517	5.23	25
MPP#2	1.0	1.38	4.9	30
MPP#3	1.0	1.43	7.54	17.5

For an overall SPL L_P, the incident pressure $\overline{P}_i$ in the present model is calculated using the reflected pressure coefficient R_f on the surface of the panel which is calculated by iteration procedure [38] using Eq. (25).

$$\overline{P}_i = \frac{P_{ref}10^{L_P/20}}{\left|1 + R_f\right|}. \tag{32}$$

Three MPPs whose parameters are summarized in Table 1 are manufactured and tested. The experimental measurements for MPP #1 are made at 125 dB and 150 dB and the normal incidence sound absorption coefficients are shown in Fig. 11. The results of the present model are in good agreement with the measurements over the entire frequency range. In Figs. 12 and 13, the normal incidence sound absorption coefficients predicted by the present model are compared with measurements performed at 140 dB and 150 dB for MPP #2. Again the results are in good agreements, especially for the high SPL case. Fig. 14 shows the estimated sound absorption coefficients by the present impedance model and the measurement results of MPP #3 at 150 dB. Excellent agreement is observed.

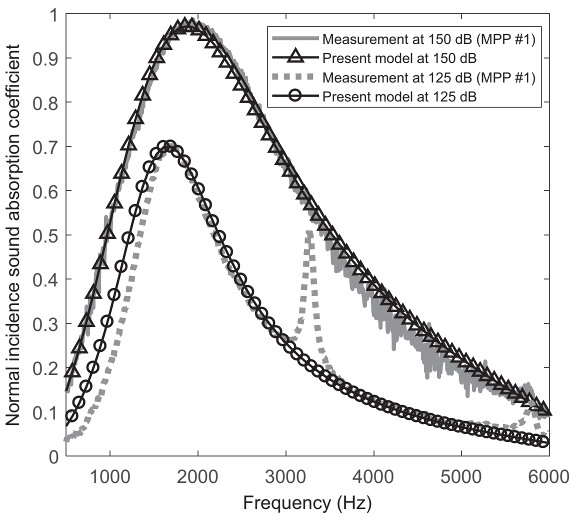

Fig. 11. Comparison of measured and estimated normal incidence absorption coefficient of MPP #1 (thickness = 0.86 mm, hole diameter = 1.517 mm, POA = 5.23%, depth of cavity = 25 mm) at 125 dB and 150 dB.

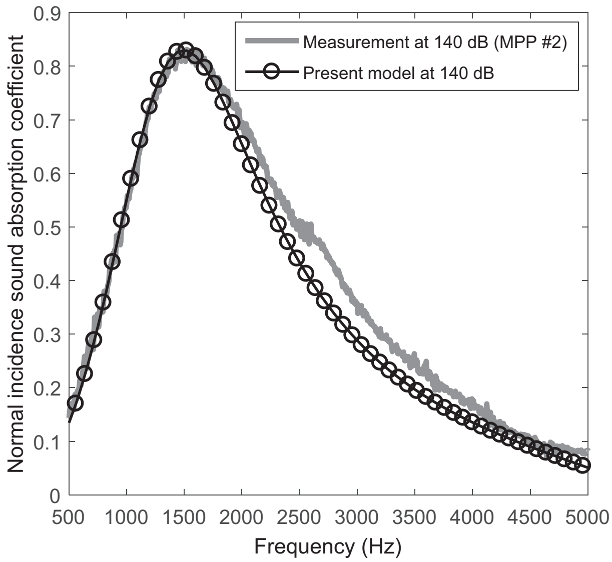

Fig. 12. Comparison of measured and estimated normal incidence absorption coefficient of MPP #2 (thickness = 1.0 mm, hole diameter = 1.38 mm, POA = 4.9%, depth of cavity = 30 mm) at 140 dB.

Figs. 11–14 demonstrate that the present impedance model is in good agreement with the experimental results. The second absorption peak in the measured results of MPP #1 at 125 dB around 3200 Hz in Fig. 11 is due to the vibration of the panel coupled to the air cavity. The resonant fundamental frequency of the plate vibration coupled with air cavity depends on the parameters of the plate and the cavity depth.

A double layer MPP absorber illustrated in Fig. 15 and constituted by MPP_1 (thickness = 1.0 mm, hole diameter = 1.4 mm, perforation ratio = 6.29%) and MPP_2 (thickness = 1.0 mm, hole diameter = 1.33 mm, perforation ratio = 4.65%) with two same cavity depths of 30 mm is manufactured and tested at 145 dB. The surface impedance of the double layer MPP absorber is obtained by transfer matrix method which can be used to determine the SPL on the surface of the second MPP. An identical mean incident pressure level is considered on the surface of both MPPs.

The predicted normal incidence sound absorption coefficient by the present model and the measurement results are presented in Fig. 16. Again, the proposed model agrees well with the measurement.

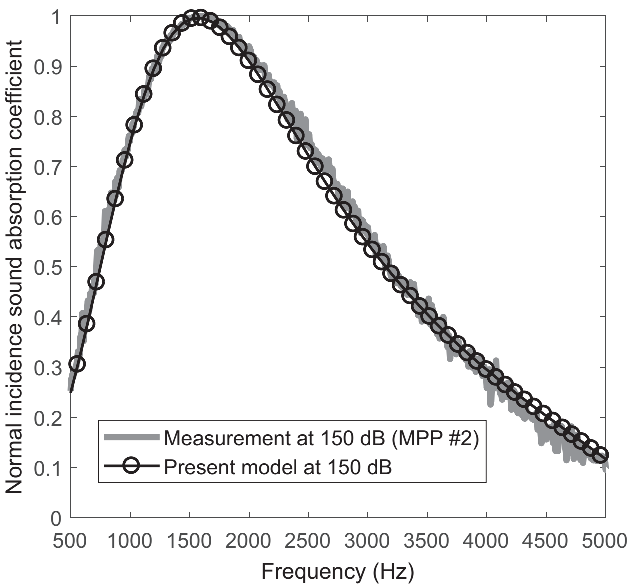

Fig. 13. Comparison of measured and estimated normal incidence absorption coefficient of MPP # 2 (thickness = 1.0 mm, hole diameter = 1.38 mm, POA = 4.9%, depth of cavity = 30 mm) at 150 dB.

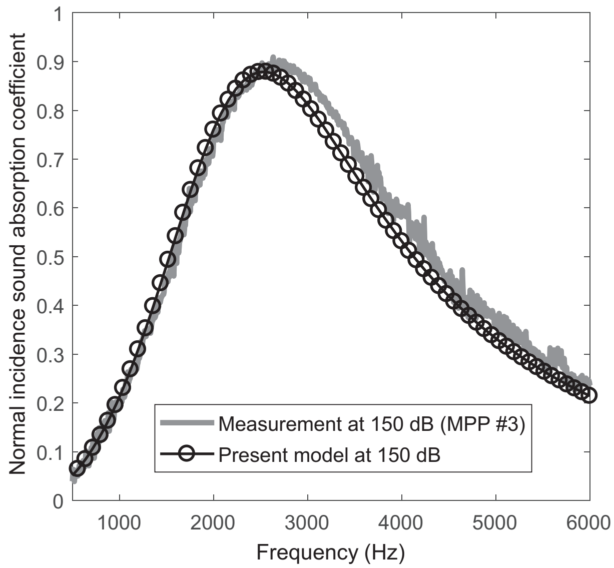

Fig. 14. Comparison of measured and estimated normal incidence absorption coefficient of MPP #3 (thickness = 1.0 mm, hole diameter = 1.43 mm, POA = 7.54%, depth of cavity = 17.5 mm) at 150 dB.

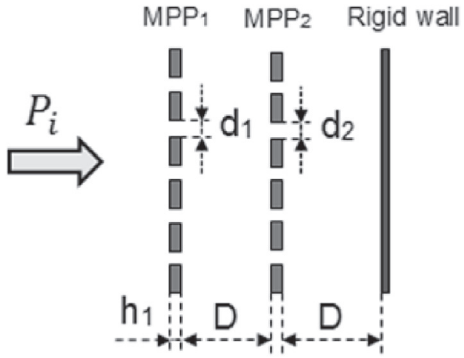

Fig. 15. Double layer micro-perforated panel absorber.

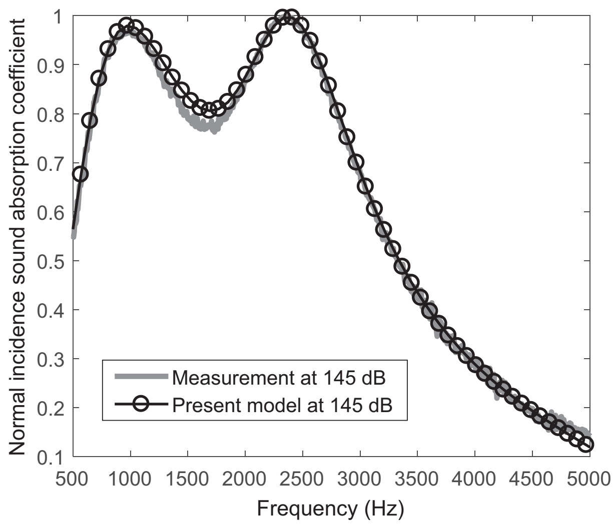

Fig. 16. Normal incidence sound absorption coefficient of a double layer MPP absorber.

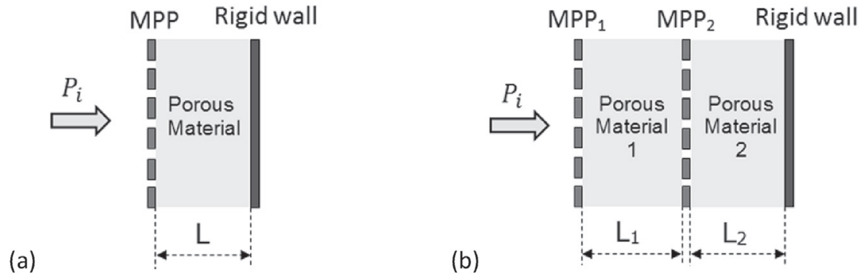

Fig. 17. MPP in contact with porous materials.

6. MPP in contact with porous mediums

In this session, micro-perforated panels backed by porous mediums as shown in Fig. 17 are modeled by equivalent fluid approach at high SPL and tested experimentally.

In Fig. 17(a), the MPP is in contact with porous material backed by rigid wall while in Fig. 17(b), MPP_2 separates two porous mediums. The acoustic properties of MPPs in both configurations are affected by the porous materials which are modeled using equivalent fluid method proposed by Johnson-Champoux-Allard [3] for porous media having rigid frame. Preliminary tests presented in Figs. 18 and 19 show that the normalized acoustic impedance and the sound absorption coefficient of the porous material are not too sensitive to the SPL in the frequency range of interest (a function of the selected MPP). Note however, that at low frequencies (f < 1000 Hz), Fig. 18 shows that the normalized resistance increases slightly with the SPL while the reactance is invariable in the entire frequency range. The main effect of the MPPs is assumed to be the distortion of the flow through the porous materials which involves an inertial and resistive effect. This distortion is accounted for by using an addition of a reactance term which depends on the dynamic tortuosity of the porous layers. In this work, equivalent tortuosity is proposed at high SPL for the micro-perforated panels following Atalla and Sgard [1] to predict the acoustic behavior of the absorbers illustrated in Fig. 17. A correction term is presented in the expression of the equivalent tortuosity which depends on the real part of the dynamic tortuosity $\mathrm{Re}(\tilde{\alpha}_p)$ of the porous layer and is expressed as $j\omega\mathrm{Re}(\tilde{\rho}_p)\varepsilon_{enl}$ with $\tilde{\rho}_p = \rho_0\tilde{\alpha}_p$ the effective density of the porous material. The correction length ε_{enl} associated to the radiation in air (Eq. (22)) is reduced by the higher pressure levels. It is a function of the particle acoustic velocity in the perforation. Then, the equivalent tortuosity of the MPP in Fig. 17(a) using the correction term is given by

$$\alpha_{\infty nl} = 1 + \frac{\varepsilon_{enl}}{h}\left(1 + \mathrm{Re}(\tilde{\alpha}_p)\right). \tag{33}$$

In Fig. 17(b), the equivalent tortuosity of MPP_1 is determined by Eq. (33) using the thickness h_1 of MPP_1 and the dynamic tortuosity $\tilde{\alpha}_{p_1}$ of the porous material 1. The equivalent tortuosity of MPP_2 (Fig. 17(b)) is given by

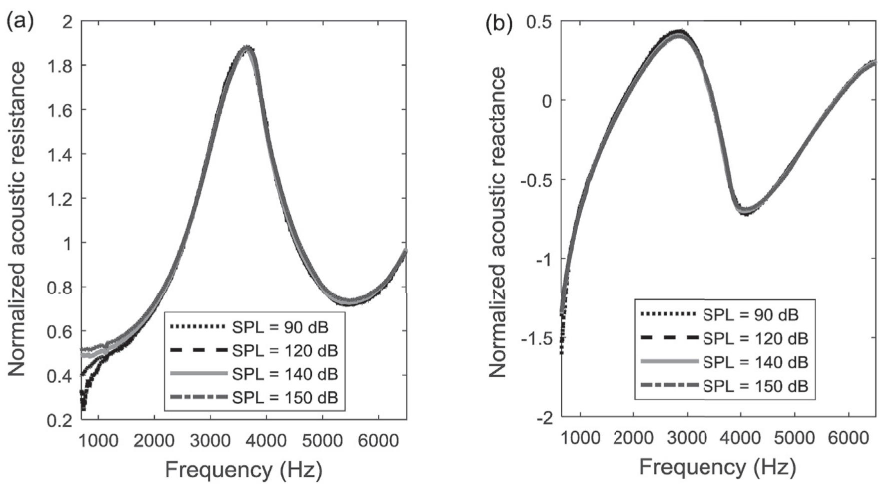

Fig. 18. Normalized surface impedance of the porous material for various SPL.

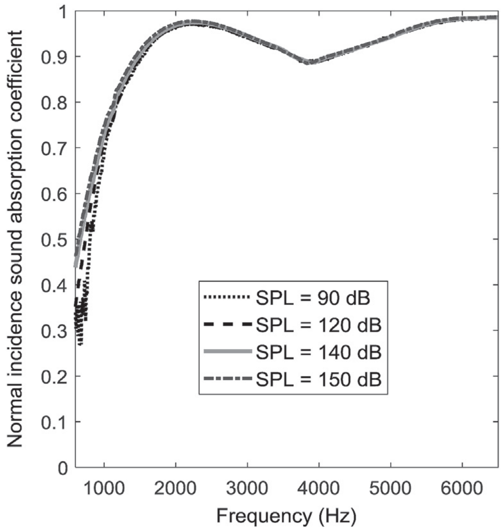

Fig. 19. Normal incidence sound absorption coefficient of the porous material for various SPL.

$$\alpha_{\infty nl} = 1 + \frac{\varepsilon_{ed}}{h_2}\left(\mathrm{Re}\left(\tilde{\alpha}_{p_1}\right) + \mathrm{Re}\left(\tilde{\alpha}_{p_2}\right)\right) \tag{34}$$

where h_2 is the thickness of MPP_2 and $\tilde{\alpha}_{p_2}$ the dynamic tortuosity of the porous material 2. The correction length ε_{ed} is given by Eq. (22) using the incident SPL on the surface of MPP_2 which can be calculated by transfer matrix method knowing the incident SPL on the surface of MPP_1. The incident SPL on the surface of MPP_2 will be lower than the one on the surface of MPP_1 because of the acoustic energy dissipation created by the porous material 1. It will generally be in the linear range according to the length and the acoustic properties of porous material 1.

The acoustic impedance on the surface of the MPP in Fig. 17(a) is expressed as

$$Z_s = Z_{MPP} + Z_P, \tag{35}$$

where Z_P is the normal impedance in front of the porous material, just at the rear of the MPP and is given by

$$Z_P = -j\frac{Z_c}{\phi_p}\cot(k_c L), \tag{36}$$

with ϕ_p the porosity of the porous layer, L is the length, Z_c and k_c are respectively the characteristic impedance and the wavenumber in the porous material (Fig. 17(a)). The impedance Z_{MPP} of MPP is calculated using the expression in Eq. (24)

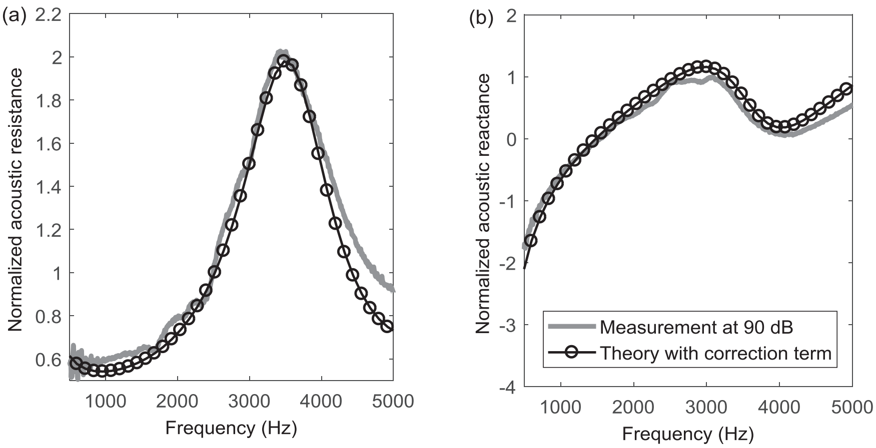

Fig. 20. Normalized surface impedance of MPP backed by porous material at 90 dB: (a) normalized resistance, (b) normalized reactance.

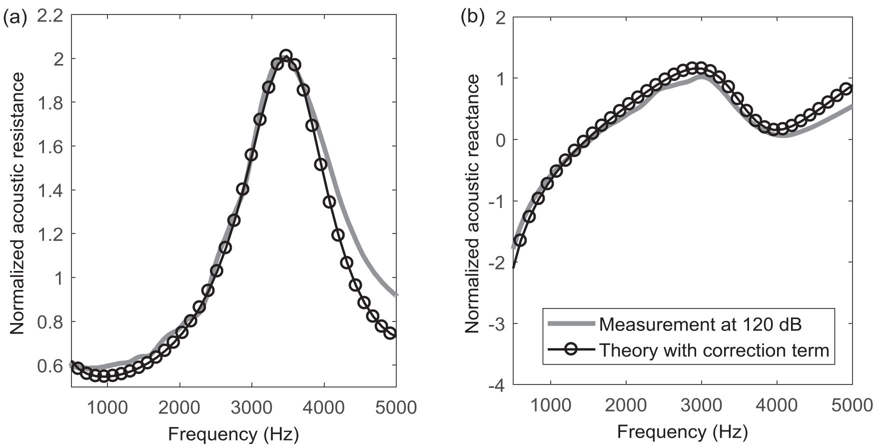

Fig. 21. Normalized surface impedance of MPP backed by porous material at 120 dB: (a) normalized resistance, (b) normalized reactance.

where the tortuosity is substituted by the relation in Eq. (33). The acoustic properties of the absorber in Fig. 17(b) are obtained by transfer matrix method. Each porous media can be modeled by the matrix

$$M_i = \begin{bmatrix} \cos(k_{c_i}L_i) & j\dfrac{\omega\rho_i}{k_{c_i}}\sin(k_{c_i}L_i) \\ j\dfrac{k_{c_i}}{\omega\rho_i}\sin(k_{c_i}L_i) & \cos(k_{c_i}L_i) \end{bmatrix}. \tag{37}$$

where $i = 1, 2$, ρ_i and L_i are respectively the density and the length of the porous layers and k_{c_i} the wavenumber. The transfer matrix associated to each MPP is written as $N_i = \begin{bmatrix} 1 & Z_{MPPi} \\ 0 & 1 \end{bmatrix}$ with Z_{MPPi} the acoustic impedance (Eq. (24)). The global matrix of the absorber (Fig. 17(b)) is then given by $T = N_1M_1N_2M_2$ which is used to obtain the surface impedance of the absorber.

A MPP with thickness of 1.0 mm, hole diameter 1.43 mm and POA 13.6% is backed by porous material (Fig. 17(a)) and tested at 90 dB, 120 dB and 130 dB. The measurement results are compared with theoretical results using the correction term given in Eq. (33). The flow resistivity of the porous material is 12000 N s m^{-4}, the porosity is 98%, the tortuosity is 1.08, the characteristic viscous and thermal lengths are respectively 135 μ m and 380 μ m, the thickness of the porous layer is 42 mm.

The normalized surface impedance and the normal sound absorption coefficients predicted by the present models using the correction term and the measurements results are presented in Figs. 20–23. The results of the existing models are not presented in this session because they are not suitable for a good prediction of the acoustic properties of MPP backed by porous material since they do not consider the correction made in the present work to take into account the effect of the

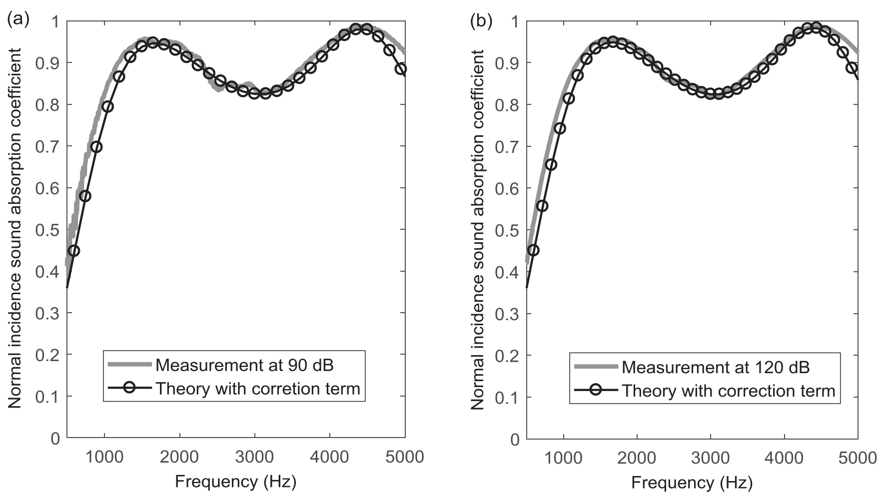

Fig. 22. Normal incidence sound absorption coefficient of MPP backed by porous material: (a) 90 dB, (b) 120 dB.

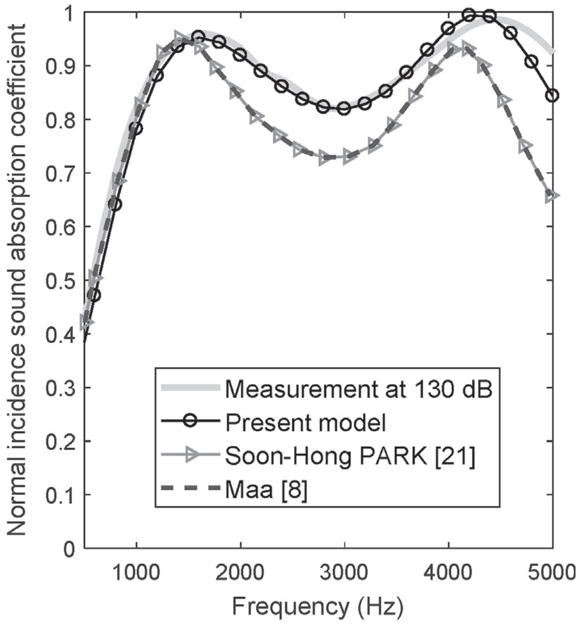

Fig. 23. Normal incidence sound absorption coefficient of MPP backed by porous material at 130 dB.

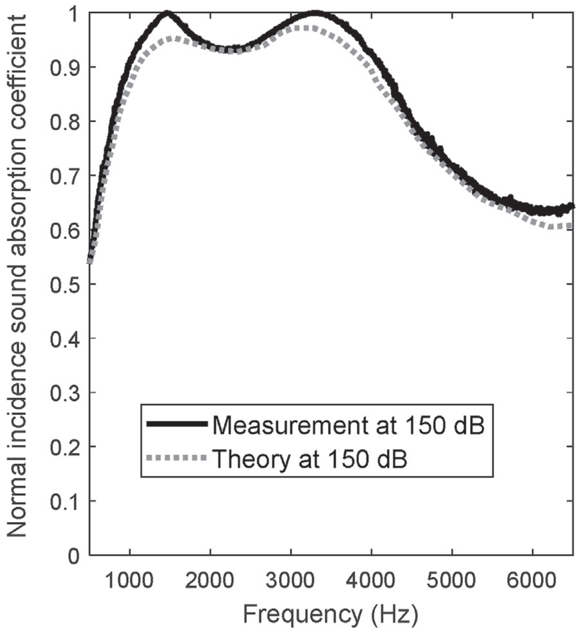

Fig. 24. Normal incidence sound absorption coefficient of MPPs backed by porous materials at 150 dB.

distortion induced by the perforations through the porous material. For completeness the results using the models of Maa [8] and Park [21] are shown in Fig. 23 to stress the importance of the tortuosity correction used in our model.

In Figs. 20 and 21, the normalized acoustic resistance and reactance of the MPP backed by porous material, respectively at 90 dB and 120 dB show a good agreement between theoretical results and measurements. The normalized resistance presents a peak at 3500 Hz where its value is 2. However a slight discrepancy is observed for frequencies higher than 4200 Hz.

The theoretical results and the measured one at 90 dB, 120 dB and 130 dB in Figs. 22 and 23 show a good correlation for the sound absorption coefficient. The sound absorption frequency band is large and two maximum absorption peaks are observed respectively at 1500 Hz and 4500 Hz. The predicted results using the models of Park [21] and Maa [8] underestimate the

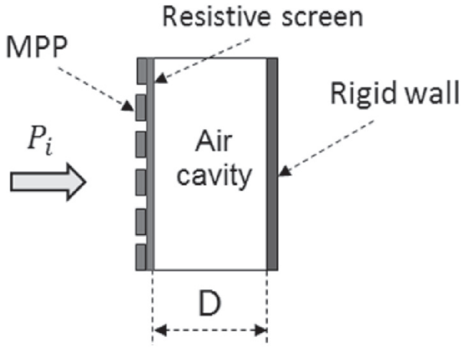

Fig. 25. Micro-perforated panel backed by resistive screen.

sound absorption coefficient in Fig. 23 and this demonstrates that a correction to the equivalent tortuosity is necessary when a porous material backs the MPP.

Fig. 24 illustrates the comparison of the sound absorption coefficient for a MPP absorber in Fig. 17(b) at 150 dB. The theoretical result is predicted using the transfer matrix method and the impedance model in Eq. (24) with the equivalent tortuosity in Eqs. (33) and (34). MPP_1 (thickness = 1.0 mm, hole diameter = 1.43 mm, perforation ratio = 13.6%) and MPP_2 (thickness = 0.88 mm, hole diameter = 1.5 mm, perforation ratio = 5.3%) are backed by the previous porous material with thickness of 20 mm.

The present model predicts well the sound absorption coefficient (Fig. 24) using the proposed correction terms (Eqs. (33) and (34)).

A last validation case is presented in Fig. 25 where the MPP (thickness = 1.0 mm, hole diameter = 1.46 mm, perforation ratio = 13.8%) is backed by a resistive screen coupled to air cavity. It is shown that the resistive screen has a linear behavior which is not affected by the SPL [40].

The MPP is characterized by the present impedance model (Eq. (24)) using Eq. (33) while the linear equivalent fluid model [1] is used for the screen. The resistance per unit area of the screen is 0.8 with a thickness of 0.16 mm, the cavity depth is 18 mm and the surface impedance of the absorber is obtained by transfer matrix method. The comparison of the theoretical results with the experiments at 150 dB is presented in Figs. 26 and 27 for the surface impedance and the sound absorption coefficient.

The theoretical results and the experimental ones are in good agreement. The normalized reactance is zero at 2700 Hz (Fig. 26) which corresponds to the resonance of the sound absorption. The absorption frequency band (Fig. 27) is larger than the ones of classical MPP absorbers in Figs. 11–14.

The present acoustic impedance model by the equivalent fluid method is then able to characterize MPP backed by porous media. The tortuosity of the MPP is corrected to account for the distortion of the flow induced by the perforations through the porous material. The proposed correction term depends on the incident sound pressure on the surface of MPP and the dynamic tortuosity of the porous layer. The existing models don't consider the correction made in this work to account for the

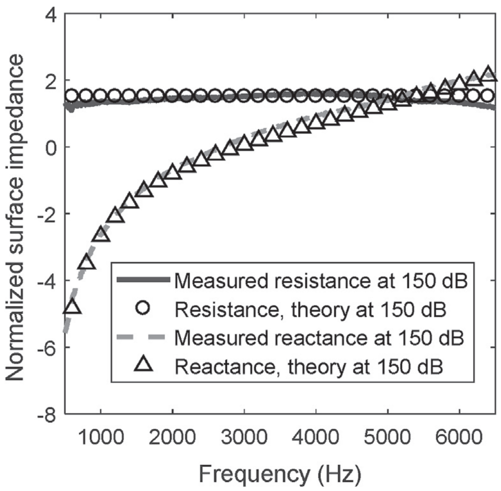

Fig. 26. Normalized surface impedance of MPP backed by resistive screen at 150 dB.

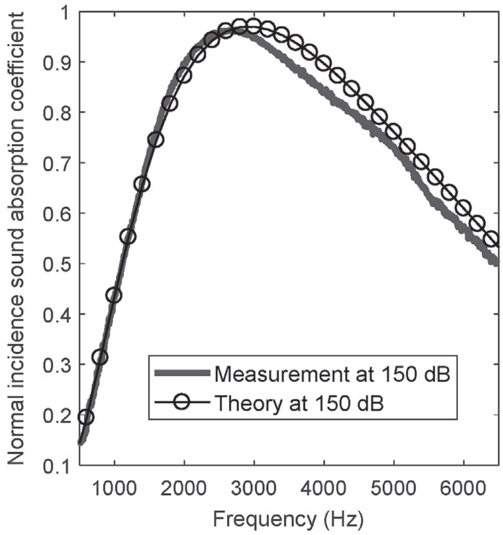

Fig. 27. Normal incidence sound absorption coefficient of MPP backed by resistive screen at 150 dB.

effect of the flow distortion caused by the perforations in the porous material; they are not thus suitable for predicting correctly the acoustic response of MPP backed by porous media.

7. Effect of the high SPL

Finally a parameters study, using the proposed model is presented to illustrate the effect of high SPL on the acoustic properties of MPP. The high SPL effect on the tortuosity of MPP which expression is given in Eq. (28) is shown in Fig. 28. The thickness of the MPP is 1.0 mm, and the hole diameter is 0.8 mm.

For low SPL, the tortuosity of the MPP decreases slightly with respect to the POA. When the pressure level exceeds 100 dB, the tortuosity becomes lower and tends to unity for high SPL. This is due to the nonlinear effects which decrease the correction length [27].

The flow resistivity of the MPP (thickness = 1.2 mm, hole diameter = 0.8 mm, POA = 1.8%) is shown in Fig. 29 with respect to the SPL by using the proposed expression in Eq. (27).

The flow resistivity of the MPP in Fig. 29 is constant for a SPL below 110 dB. The increase of the SPL induces a high flow resistivity of the MPP.

In Figs. 30 and 31, the normal incidence sound absorption coefficients of a MPP absorber are presented with respect to the frequency and the perforation diameter at 100 dB and 135 dB respectively. The thickness of the MPP is 1.0 mm, the perforation ratio is 5.3% and the cavity depth is 30 mm.

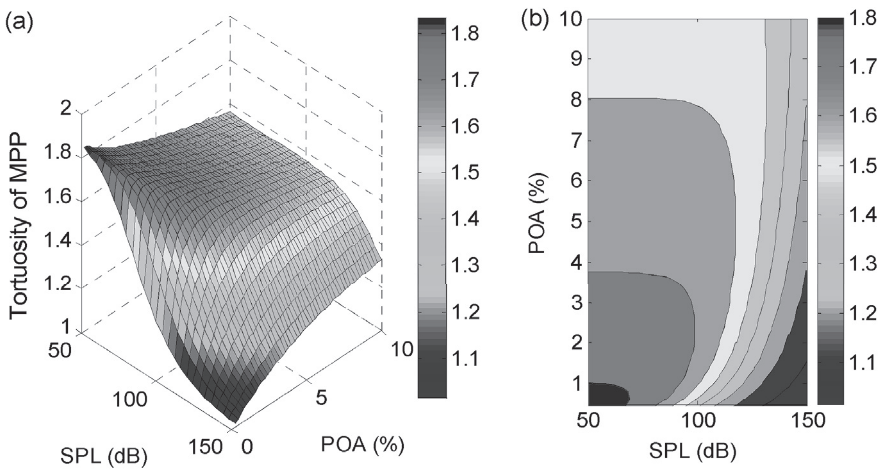

Fig. 28. The tortuosity of the MPP (thickness = 1.0 mm, hole diameter = 0.8 mm): (a) surface plot, (b) contour plot.

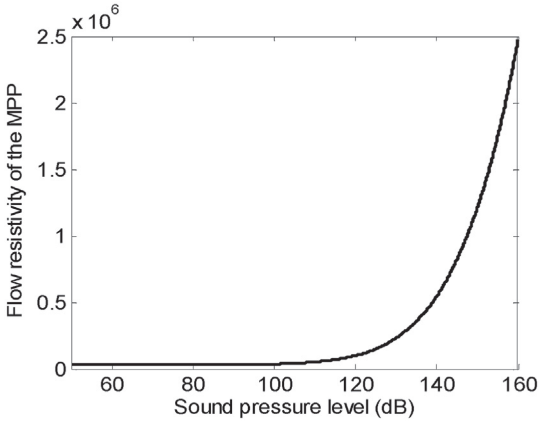

Fig. 29. The flow resistivity of the MPP (thickness = 1.2 mm, hole diameter = 0.8 mm, POA = 1.8%).

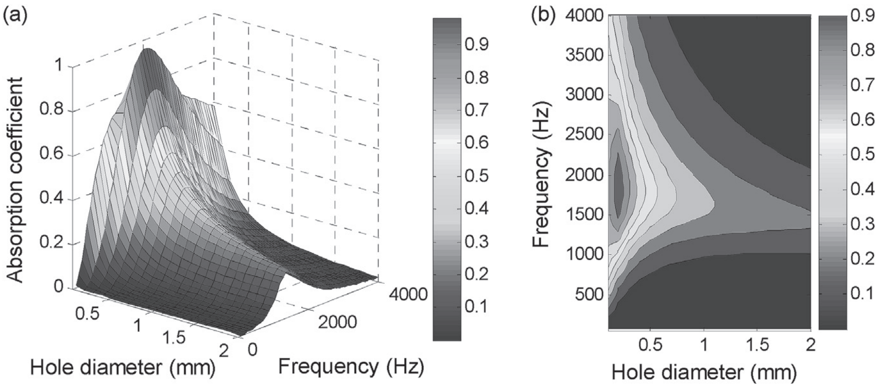

Fig. 30. Normal incidence sound absorption coefficient of MPP absorber (SPL = 100 dB, thickness = 1.0 mm, POA = 5.3%, depth of cavity = 30 mm): (a) surface plot, (b) contour plot.

The sound absorption at 100 dB in Fig. 30 is more interesting when the hole diameter is smaller than unity. Indeed, for low SPL, when the dimensions of the perforations are in the order of viscous and thermal boundary layers thicknesses, the acoustic energy is converted into heat through friction by viscous and thermal effects. The thickness of the viscous boundary layer depends on the frequency and the viscous properties of the fluid and is given by Refs. [1,37] $d_{vis} = \sqrt{2\eta/(\omega\rho_0)}$ and is between 0.015 mm and 0.5 mm for air at audible frequency (20 Hz-20 kHz) under normal conditions. The thermal boundary

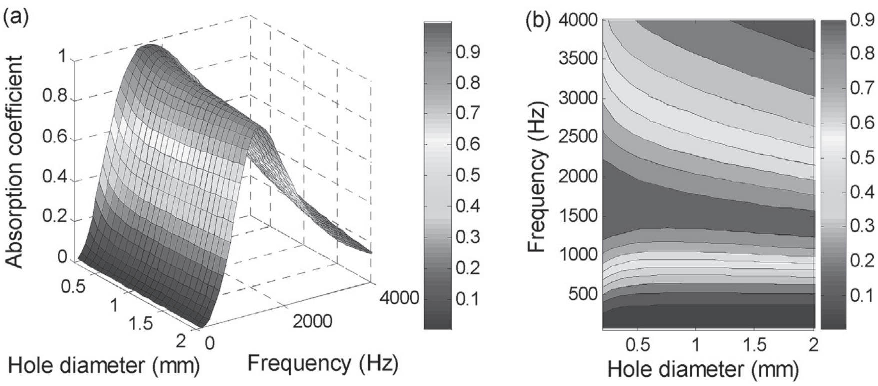

Fig. 31. Normal incidence sound absorption coefficient of MPP absorber (SPL = 135 dB, thickness = 1.0 mm, POA = 5.3%, depth of cavity = 30 mm): (a) surface plot, (b) contour plot.

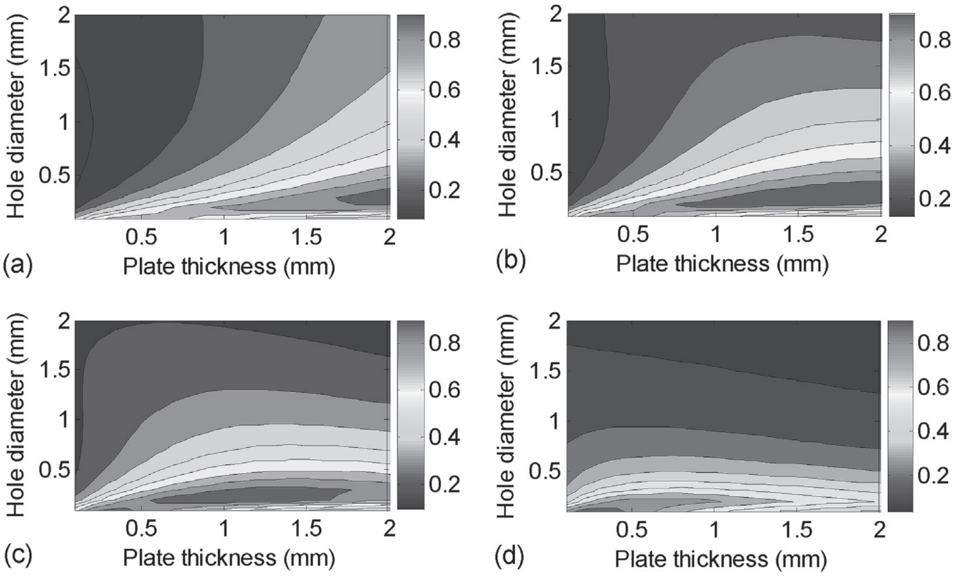

Fig. 32. Normal incidence sound absorption coefficient of MPP absorber (SPL = 100 dB, POA = 3.8%, depth of cavity = 40 mm): (a) f = 800 Hz, (b) f = 1000 Hz, (c) f = 1200 Hz, (d) f = 1800 Hz.

layer thickness given by $d_{th} = \sqrt{\lambda/(\omega\rho_0 C_V)}$ with λ the thermal conduction, C_V the specific heat (per unit mass) at constant volume, is between 0.02 mm and 0.58 mm for air. Beyond the boundary layers thicknesses, the acoustic dissipation in the linear regime is lower and the MPP absorber is not efficient in sound absorption (Fig. 30). Thus, at low SPL, MPP absorbers with a large hole diameter and high perforation ratio are poor sound absorbers. In Fig. 31, a MPP absorber with a large perforation diameter, which shows a poor sound absorption coefficient at low SPL (Fig. 30), presents a good absorption coefficient at 135 dB. At high SPL, as explained in Ref. [10], the flow separation appears at the exit of the orifice in the form of a high velocity jet and the acoustic energy is dissipated by conversion into kinetic energy of the vortices. A MPP absorber with a large perforation diameter can thus be effective in sound absorption in the nonlinear regime.

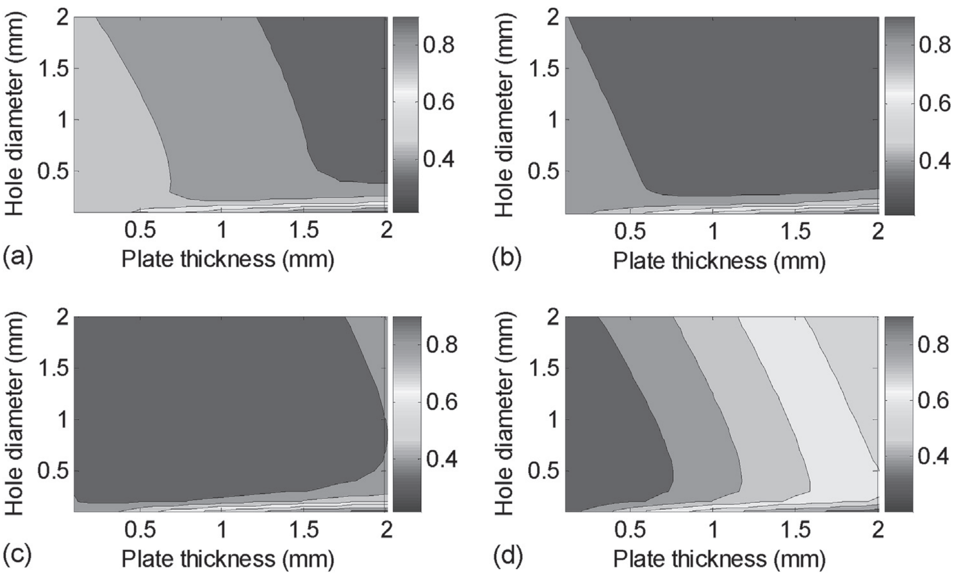

Fig. 33. Normal incidence sound absorption coefficient of MPP absorber (SPL = 140 dB, POA = 3.8%, depth of cavity = 40 mm): (a) f = 800 Hz, (b) f = 1000 Hz, (c) f = 1200 Hz, (d) f = 1800 Hz.

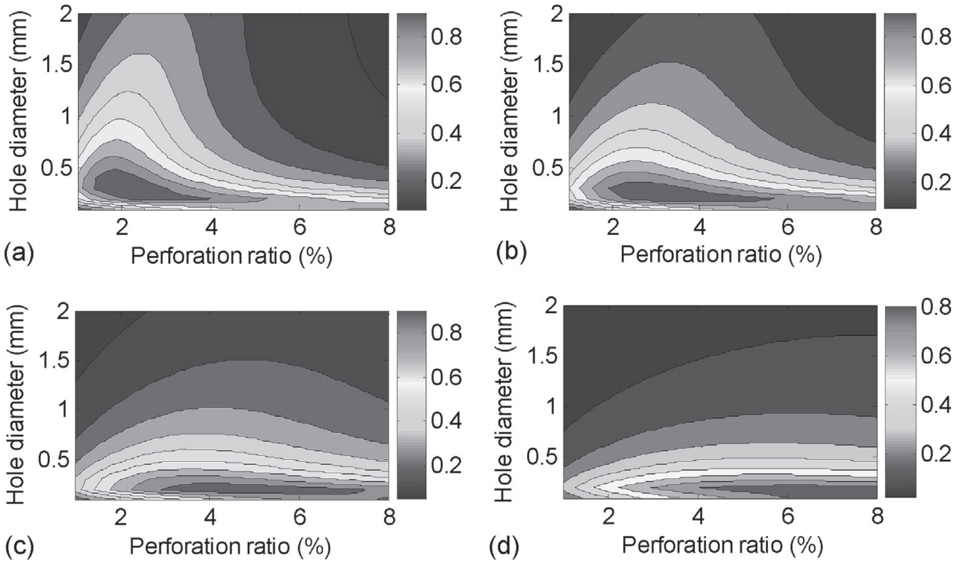

Fig. 34. Normal incidence sound absorption coefficient of MPP absorber (SPL = 100 dB, thickness = 1.0 mm, depth of cavity = 35 mm): (a) f = 1000 Hz, (b) f = 1200 Hz, (c) f = 1500 Hz, (d) f = 2000 Hz.

The range of the perforation diameters and the plate thicknesses to obtain a maximum sound absorption coefficient at each frequency is illustrated at 100 dB in Fig. 32 and 140 dB in Fig. 33 for selected frequencies: 800 Hz, 1000 Hz, 1200 Hz and 1800 Hz. The illustration is performed for a MPP absorber with perforation ratio of 3.8% and cavity depth of 40 mm.

In Fig. 32 the maximum of the sound absorption coefficient at 100 dB is obtained for a perforation diameter smaller than 1 mm. As explained previously, the MPP absorber cannot be efficient in sound absorption in the linear regime with a large hole diameter. In Fig. 33, the range of the plate thickness and perforation diameter to get a maximum absorption at 140 dB is large. With a hole diameter greater than 1 mm, the sound absorption coefficient is up to 0.9 at 1000 Hz and 1200 Hz in Fig. 33(b) and (c).

Finally, to illustrate the range of orifice diameters and perforation ratio for maximum absorption coefficient, the contour plots at 100 dB and 143 dB of the sound absorption coefficient of a MPP absorber with thickness of 1.0 mm and cavity depth of 35 mm are presented in Figs. 34 and 35, respectively, for selected frequencies: 1000 Hz, 1200 Hz, 1500 Hz and 2000 Hz. Again,

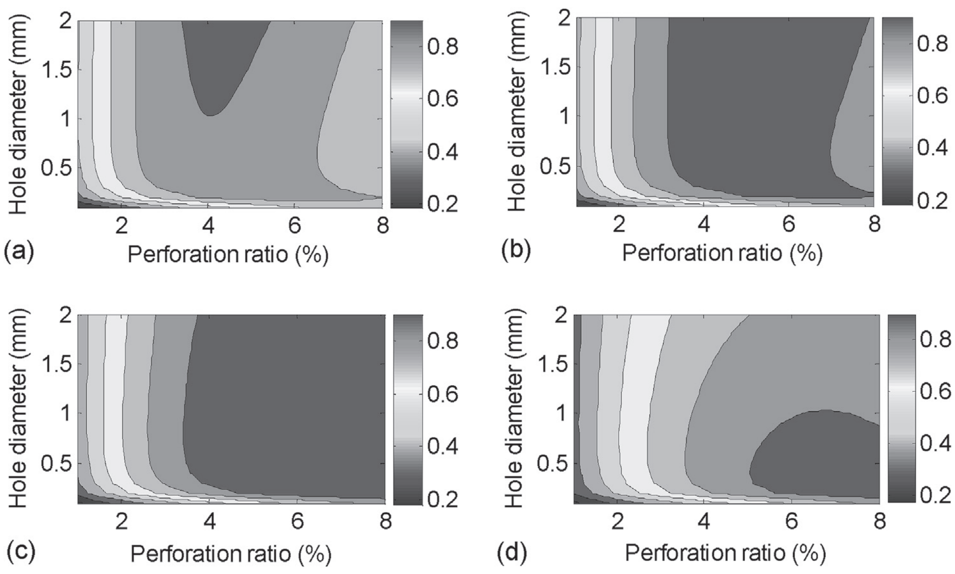

Fig. 35. Normal incidence sound absorption coefficient of MPP absorber (SPL = 143 dB, thickness = 1.0 mm, depth of cavity = 35 mm): (a) f = 1000 Hz, (b) f = 1200 Hz, (c) f = 1500 Hz, (d) f = 2000 Hz.

it is clearly observed that the range of the hole diameter and perforation ratio to get a maximum absorption is limited in the linear regime (Fig. 34) while it is much more broader in the non-linear regime (Fig. 35).

8. Conclusion

An acoustic impedance model of a MPP in high sound pressure environment is proposed. The MPP is modeled as an equivalent fluid using Johnson-Allard approach for rigid porous media with an effective density. Unlike existing models which are case specific and generally limited to micro-perforated panels backed by an air cavity, the present model predicts correctly at high SPL the acoustic response of a micro-perforated panel and couples naturally with various domains such as an air cavity, a porous material or a resistive screen. For the latter cases, a correction to the MPP equivalent tortuosity was proposed to account for the effect of the flow distortion caused by the perforations through the porous media. This correction which depends on the dynamic tortuosity of the porous media and the incident sound pressure on the surface of the perforations is not done in the existing models. In all configurations where MPP is backed by air cavity, porous material or resistive screen, the proposed model has demonstrated a good correlation with the experimental results. The present nonlinear impedance model can be used to predict the acoustic behavior of MPP at high SPL. It is shown that a MPP absorber with a large perforation diameter which is poor in sound absorption in the linear regime can become a good sound absorber in the nonlinear regime.

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